
Two New Text-Tabular Datasets for MulTaBench

Noa Bamberger

Technion - Israel Institute of Technology
NLP (00970215) - Track 2 (Benchmark Track)
noabamberger@campus.technion.ac.il

Abstract

MulTaBench is a dedicated benchmark for multimodal tabular learning with text and image modalities. Expanding it with qualifying datasets broadens domain diversity and supports more robust, generalizable evaluations of multimodal models. Using the official MulTaBench curation protocol, we screened 232 candidate datasets through an automated pipeline, fully evaluated five candidates across five tabular learners (TabM, CatBoost, LightGBM, TabPFN v2, and TabPFN-2.5) under four experimental conditions, and identified two new text-tabular datasets that satisfied the acceptance criterion. Both datasets demonstrate substantial multimodal improvements over unimodal baselines. *Vietnam housing* qualifies across all five learners, exhibiting strong joint signal and statistically significant target-aware representation (TAR) gains. *Udemy courses* also satisfy the benchmark acceptance threshold on a majority of learners (3 of 5), though its TAR effect does not reach statistical significance when pooled across folds. Finally, we detail the automated mining pipeline used to screen candidates and discuss two common screening shortcuts that we assess to be invalid.

1 Introduction

Real tables are rarely purely numerical: a property listing carries an address, a course listing a title and an instructor name - free text beside numeric and categorical columns. *Text-tabular learning* asks how to use both channels at once. The question is not rhetorical in that gradient-boosted trees remain very strong on structured data [1, 2]. Accordingly, a multimodal method only earns its complexity on tables where the text carries signal the structured block lacks, and vice versa. This makes *dataset curation* a substantive research task rather than a simple data collection exercise. A benchmark built from tables whose text merely restates their columns will report that multimodal methods offer little to no value added. MulTaBench [3] makes the requirement measurable in that a table is worth benchmarking only if (i) the joint model beats *both* unimodal baselines, and (ii) a text encoder adapted to the prediction target beats a frozen one. Failing (i) collapses the task into pure tabular or pure NLP. Failing (ii) exercises text encoding but not the target-aware representation learning that models such as TabSTAR [4] are built around. This raises the practical question of how to identify, from a large pool of candidate datasets, those tables that actually satisfy these requirements.

Our contributions are therefore threefold: (1) **The identification of two new text-tabular datasets** that qualify under the MulTaBench curation protocol: Vietnamese residential real-estate listings [5] (accepted across 5 of 5 tabular learners) and online-course listings [6] (accepted across 3 of 5), both demonstrating consistent multimodal gains over unimodal baselines [Section 4]. (2) **The description of an end-to-end automated mining pipeline** for discovery, column typing, and leakage screening [section 3]. From an initial pool of 232 screened candidates, resource constraints limited full-grid evaluation to a small subset, from which these two datasets were validated [section 4]. We report the survival rates across each filtering stage to guide future data collection efforts [section 3].

(3) **The explication of methodological insights into multimodal tabular curation**, demonstrating why common heuristic shortcuts - specifically, filtering structured columns by name and attempting to screen target awareness on reduced epochs or single folds - are unreliable and lead to invalid curation verdicts [section 4].

2 Related Work

Tabular learning. Gradient-boosted decision trees - XGBoost [7], LightGBM [8] and CatBoost [9] - are the default for tabular prediction and remain competitive against deep architectures [1, 2]. Deep tabular models have closed much of the gap: FT-Transformer [10] adapted attention to heterogeneous columns, and TabM [11] reaches GBDT-level accuracy through parameter-efficient ensembling of an MLP backbone. We use LightGBM, CatBoost and TabM.

Tabular foundation models. TabPFN v2 [12] performs in-context learning over a synthetic prior, strong on small tables without per-dataset gradient training; TabPFN-2.5 [13] extends it to larger inputs. Pretrained rather than fitted, they respond differently to input representation changes than do trees.

Text and tables together. One line of work serialises rows into sentences for a language model (TabLLM [14]); another embeds string entries as entities to transfer across tables (CARTE [15]). The line we follow keeps the tabular learner and encodes text columns into features with a contrastive sentence encoder, E5 [16], optionally adapted to the target. TabSTAR [4] makes that adaptation central - *semantically target-aware* representations - and MulTaBench operationalises it as the `ft` condition: an E5 encoder LoRA-tuned [17] on the target.

Benchmarks and curation. MulTaBench [3] is, to our knowledge, the first multimodal-tabular benchmark to publish an executable acceptance rule for its own datasets rather than a curated list alone. We reuse its reference implementation of that rule verbatim (Section 3), so our verdicts and the benchmark’s are one computation.

3 Method: the criterion, the pipeline, and the datasets

The criterion. Each dataset is evaluated with five learners - TabM, CatBoost, LightGBM, TabPFN v2 and TabPFN-2.5 - under four conditions: **Unimodal Structured** (tabular columns only), **Unimodal Unstructured** (text only), **Joint Frozen** (all features, frozen E5-small-v2 embeddings) and **Joint Target-Aware** (TAR; all features, E5 LoRA-tuned on the target). Text columns are embedded per column and compressed to 30 principal components; scores are R^2 , as both our datasets are regression tasks. Writing μ_c for the mean score over folds 0-4 in condition c , rounded to three decimals before differencing, a learner passes when

$$\Delta_{\text{Joint}} = \mu_{\text{frozen}} - \max(\mu_{\text{struct}}, \mu_{\text{text}}) > \delta, \quad \Delta_{\text{Aware}} = \mu_{\text{TAR}} - \mu_{\text{frozen}} > \delta, \quad (1)$$

with $\delta = 0.001$; a dataset is accepted when at least 3 of the 5 learners pass *both*, a learner clearing only one contributing nothing. Every verdict here is computed by the MulTaBench reference implementation of this rule, which asserts grid completeness (5 folds $\times$ 4 conditions per learner) rather than averaging over gaps; we did not reimplement the arithmetic. To assess whether improvements over unimodal baselines (Δ_{Joint}) and frozen representations (Δ_{Aware}) are consistent across evaluations, we also compute one-sample t -statistics testing $H_0 : \Delta_{\text{Aware/Joint}} \leq 0$ pooled over all 25 (learner, fold) evaluations.

Setup and fidelity. All grids ran on Kaggle [18] T4 GPUs, the four conditions for a learner always measured in one session so that no environment difference can enter a delta. To ensure fidelity, we validated our setup against the benchmark’s published artifacts in two steps. First, applying the curation acceptance rule to MulTaBench’s released raw scores reproduced its 56×10 pass matrix with zero mismatched cells. Second, re-evaluating a representative reference dataset (*Product Sentiment*) from scratch exactly matched published scores in the encoder-free baseline, while Δ_{Joint} agreed within 0.012 with all signs matching (Appendix H).

Table 1: Mining-funnel yield.

Stage	Count	Survival
Discovered by search	232	—
Parsed successfully	229	99%
Viable under typing/target gates	120	52%
After removing false text columns (dates, identifiers)	58	25%
After excluding domains already in MulTaBench	34	15%
Passing a screen	16	7%
Full grid attempted	5	—
Survived scrutiny	2	< 1%

Mining pipeline. Candidates were discovered through a programmatic search over public dataset repositories, then profiled, screened, and gridded (Table 1). Under our compute budget, the end-to-end yield of our current screening configuration was under 1%. Rather than brute-forcing hundreds of full grids, the pipeline focuses on maximizing candidate qualification before committing resources to complete evaluation. For each candidate, the pipeline selects a target and partitions columns without human annotation. Targets are rejected when they are identifiers or serial numbers, when one value holds more than 20% of the mass (a sentinel masquerading as a number), when $|z|_{\max} > 5$, or when derived from the text from which they would be predicted. Columns correlating with the target at Spearman $\rho \geq 0.95$ are dropped as leakage (Spearman, so monotone-but-nonlinear leaks such as a rank column are caught). Column typing is inferred, not declared, such that a non-numeric column counts as text at ≥ 100 distinct values or $\geq 80\%$ uniqueness - and it fails mainly by promoting dates and identifiers into the text block, which is what reduces 120 nominally viable candidates to 58. Two cost-saving shortcuts that look obvious are measurably invalid (Appendix F), so every positive Δ_{Joint} below was re-measured with the maximal structured block and every Δ_{Aware} at ≥ 10 epochs over all five folds.

The two datasets. Both specifications were derived using our automated pipeline:

- **Vietnam housing** (REG_TEXT_HOUSES_VIETNAM_2024; 30,229 listings) [5]: Predicts Price (687 distinct values, $|z|_{\max} = 2.54$). Features include one free-form text column (Address) paired with six numeric and four categorical physical attributes (e.g., area, floors, bedrooms), none of which duplicate the address.
- **Udemy courses** (REG_TEXT_EDU_UDEMY_ACADEMY; 2,999 listings) [6]: Predicts course price (28 distinct values, $|z|_{\max} = 3.00$). Features include two text columns (course_name, course_instr) paired with six numeric course statistics.

4 Results

Vietnam housing – accepted, 5 of 5. Pooled over the 25 (learner, fold) cells (Appendix B), $\Delta_{\text{Joint}} = 0.288 \pm 0.024$, positive in **25 of 25**, $t = 59.9$; $\Delta_{\text{Aware}} = 0.0065 \pm 0.0068$, positive in 22 of 25, $t = 4.7$. The joint condition nearly doubles the better unimodal baseline ($0.35 \rightarrow 0.65$). The mechanism is interpretable, and is the property we believe is worth searching for directly: the address encodes *location premium* while the structured block encodes *physical size and condition* - orthogonal drivers of price, neither derivable from the other. The baselines are also nearly balanced (0.34 vs 0.33), the fingerprint of two independently informative channels.

Udemy courses – accepted, 3 of 5. $\Delta_{\text{Joint}} = 0.175 \pm 0.046$ (Appendix C), positive in **25 of 25**, $t = 19.1$; both unimodal baselines are non-degenerate (≈ 0.30 structured, ≈ 0.21 text), so the gain is genuine complementarity rather than one modality collapsing. No course or MOOC dataset exists in MulTaBench, so the domain is genuinely new. However, here as well there are caveats to mention. Pooled over all 25 cells Δ_{Aware} is 0.0036 ± 0.0202 , $t = 0.9$ - *not* significant, and negative on two learners - though it meets the per-learner criterion on exactly the quorum of three.

Negative results. Of five candidates gridded [Table 1], three did not qualify (Figure A.1, Appendices D–E). *Board games* [19] cleared the joint condition ($+0.047$ to $+0.059$ with the structured

Table 2: **Per-learner curation results for candidate text-tabular regression datasets.** Values report mean R^2 across 5 folds ($\delta = 0.001$). A dataset is accepted if at least 3 of 5 learners satisfy both $\Delta_{\text{Joint}} > \delta$ and $\Delta_{\text{Aware}} > \delta$.

Dataset	Learner	Struct.	Text	Frozen	TAR	Δ_{Joint}	Δ_{Aware}	Verdict
Vietnam housing	CatBoost	0.341	0.322	0.632	0.636	+0.291	+0.004	pass
	LightGBM	0.342	0.310	0.592	0.607	+0.250	+0.015	pass
	TabM	0.312	0.336	0.633	0.638	+0.297	+0.005	pass
	TabPFN v2	0.342	0.340	0.646	0.647	+0.304	+0.001	pass [†]
	TabPFN-2.5	0.356	0.346	0.680	0.685	+0.324	+0.005	pass
Udemy courses	CatBoost	0.297	0.210	0.480	0.501	+0.183	+0.021	pass
	LightGBM	0.261	0.212	0.479	0.474	+0.218	-0.005	fail
	TabM	0.298	0.221	0.504	0.492	+0.206	-0.012	fail
	TabPFN v2	0.299	0.162	0.435	0.437	+0.136	+0.002	pass
	TabPFN-2.5	0.311	0.211	0.452	0.464	+0.141	+0.012	pass

[†] Knife-edge pass: rounded $\Delta_{\text{Aware}} = \delta$. Struct./Text/Frozen/TAR denote the four evaluation conditions.

block restored) but failed on target-awareness at 1 of 5. *Anime popularity* [20] likewise cleared it (+0.031 to +0.037) but failed at 0 of 5; pooled, its Δ_{Aware} is -0.0012, $t = -2.15$ - the tuned encoder is *significantly worse* than the frozen one. *MTG card prices* [21] cleared it as well (+0.053 to +0.074) but is borderline on target-awareness: 4 of 5 learners pass nominally, two only on a knife-edge, leaving 2 of 5 without them. A Metacritic table [22] was rejected at the target: 82% of rows carried a sentinel zero, so the regression was really a has-a-score indicator; screen-time rejections are listed in Appendix G. The pattern is itself a finding in that joint signal is comparatively easy - 16 of 34 screened candidates cleared it - while target-awareness eliminates most.

5 Conclusion and Summary

We contribute two new text-tabular datasets that pass the MulTaBench curation criterion - one at 5 of 5 learners with a joint gain of +0.288 R^2 positive in every measured cell, the other at 3 of 5 with +0.175 - both two orders of magnitude above the acceptance margin and both measured on complete 100-cell grids with no absent cells. Equally, we report that target-awareness rather than joint signal is what eliminates candidates, that it cannot be screened cheaply along either the epoch or the fold axis, and that a name-based structured-column filter manufactures a joint signal that does not exist. The most productive question when assessing a candidate is not whether it contains text, but whether the text describes a *different aspect of the entity* than the structured columns do. Both accepted datasets have that property, and restatement is where the joint gain vanishes.

Target-aware fine-tuning via LoRA [17] proved to be the most demanding bottleneck of this work. Beyond demanding substantial computational resources and prolonged evaluation times across learners and folds, the adaptation failed on the majority of candidate datasets, and in cases like anime popularity [20] it even actively harmed performance ($t = -2.15$). Even when fine-tuning succeeded, the observed performance gains (Δ_{Aware}) were largely marginal compared to the baseline joint gains (Δ_{Joint}). Future work should focus on improving the effectiveness of tabular text adaptation and establishing computationally cheaper, more reliable methods to fine-tune encoders without prohibitive screening overhead.

Limitations. *Epoch budget:* Although our measurements indicate that awareness gains grew with training epochs, due to the computational cost of evaluating many candidate datasets across multiple learners and folds, TAR fine-tuning was limited to 10 epochs rather than the reference default of 50 with patience 3. While allocating a larger epoch budget can allow the adapter to learn richer task-specific representations, this holds only if early stopping and regularization are carefully managed to avoid overfitting on smaller splits. Accordingly, we deem the 10-epoch estimates to be conservative, as reduced training is more likely to suppress a viable candidate than to create a false positive. *Comparability:* R^2 deltas depend on task difficulty, so the scale-free ratio of joint to best-unimodal performance ($1.85 \times$ Vietnam housing, $1.61 \times$ Udemy) transfers better across datasets.

References

- [1] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on typical tabular data? In *NeurIPS Datasets and Benchmarks Track*, 2022.
- [2] Ravid Shwartz-Ziv and Amitai Armon. Tabular data: Deep learning is not all you need. *Information Fusion*, 81:84–90, 2022.
- [3] Alan Arazi, Eilam Shapira, Shoham Grunblat, Mor Ventura, Elad Hoffer, Gioia Blayer, David Holzmüller, Lennart Purucker, Gaël Varoquaux, Frank Hutter, and Roi Reichart. MulTaBench: Benchmarking multimodal tabular learning with text and image. *arXiv preprint arXiv:2605.10616*, 2026.
- [4] Alan Arazi, Eilam Shapira, and Roi Reichart. TabSTAR: A tabular foundation model for tabular data with text fields. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [5] Tien Nhan Nguyen. Vietnam housing dataset 2024. <https://www.kaggle.com/datasets/nguyentiennhan/vietnam-housing-dataset-2024>, 2024.
- [6] Mariah Alshiekh. Udemy course dataset. <https://www.kaggle.com/datasets/mariahalshiekh/udemy-course-academy-teaching>, 2024.
- [7] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 785–794, 2016.
- [8] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [9] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [10] Yury Gorishniy, Ivan Rubachev, Valentin Khrulkov, and Artem Babenko. Revisiting deep learning models for tabular data. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [11] Yury Gorishniy, Akim Kotelnikov, and Artem Babenko. TabM: Advancing tabular deep learning with parameter-efficient ensembling. In *International Conference on Learning Representations (ICLR)*, 2025.
- [12] Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637(8045):319–326, 2025.
- [13] Léo Grinsztajn, Klemens Flöge, Oscar Key, Felix Birkel, Philipp Jund, Brendan Roof, Benjamin Jäger, Dominik Safaric, Simone Alessi, Adrian Hayler, et al. TabPFN-2.5: Advancing the state of the art in tabular foundation models. *arXiv preprint arXiv:2511.08667*, 2025.
- [14] Stefan Hegselmann, Alejandro Buendia, Hunter Lang, Monica Agrawal, Xiaoyi Jiang, and David Sontag. TabLLM: Few-shot classification of tabular data with large language models. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 5549–5581, 2023.
- [15] Myung Jun Kim, Léo Grinsztajn, and Gaël Varoquaux. CARTE: Pretraining and transfer for tabular learning. In *International Conference on Machine Learning (ICML)*, 2024.
- [16] Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533*, 2022.

- [17] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.
- [18] Kaggle. Kaggle notebooks: Free cloud service for data science and machine learning. <https://www.kaggle.com/code>, 2026. Experiments conducted using the NVIDIA T4 GPU accelerator environment.
- [19] Melissa Monfared. Board games. <https://www.kaggle.com/datasets/melissamonfared/board-games>, 2021.
- [20] Michał Błaszczuk. Ratings of the most popular anime. <https://www.kaggle.com/datasets/michau96/ratings-of-the-most-popular-anime>, 2021.
- [21] Drew Wham. Mtg card prices. <https://www.kaggle.com/c/mtgcardprices>, 2019.
- [22] The Devastator. Top pc games based on metascore (metacritic.com). <https://www.kaggle.com/datasets/thedevastator/highest-rated-pc-games-on-metacritic-metascore>, 2022.

Appendices

The appendices contain result tables and a summary figure. All reported values were computed directly from the committed run CSV files in `results/curation/` in the repository above. Scores are reported as R^2 . Per-state columns are means over folds 0-4 rounded to three decimals, matching the order of operations the criterion uses. Pooled statistics are computed on unrounded per-(learner, fold) values.

A Outcome of every gridded (dataset, learner) cell

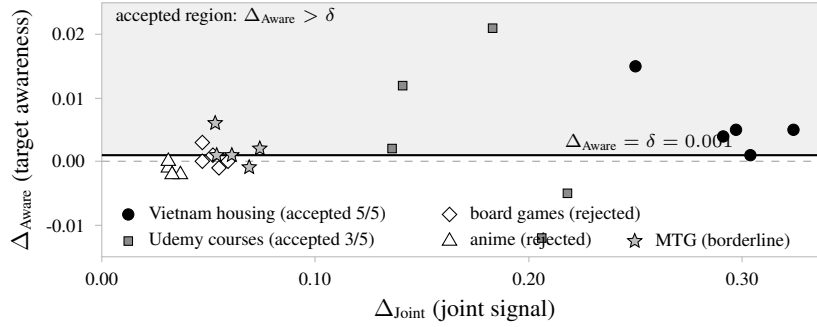


Figure A.1: Every gridded (dataset, learner) cell, Δ_{Joint} against Δ_{Aware} , from the per-learner five-fold means. A learner passes only inside the shaded band.

The horizontal spread is much larger than the vertical spread. Joint signal is relatively easy to find, while target awareness is what eliminates candidates. The three non-accepted datasets sit far to the left on Δ_{Joint} and cluster on the acceptance line on Δ_{Aware} : of their fourteen measured learners, only three sit above it, none by more than $+0.006$, while the two accepted ones clear it on both axes. Nothing in the figure separates accepted from rejected along the horizontal axis alone.

B Vietnam housing - complete T4 grid

Table B.1: REG_TEXT_HOUSES_VIETNAM_2024, 100 cells (5 learners $\times$ 4 states $\times$ 5 folds), one Tesla T4, ft at 10 epochs. Sources: `grid_frozen_gpu.csv`, `grid_tar.csv`, `grid_tar_pfn25.csv`.

Learner	Struct.	Text	Frozen	TAR	Δ_{Joint}	Δ_{Aware}	Verdict
CatBoost	0.341	0.322	0.632	0.636	+0.291	+0.004	pass
LightGBM	0.342	0.310	0.592	0.607	+0.250	+0.015	pass
TabM	0.312	0.336	0.633	0.638	+0.297	+0.005	pass
TabPFN v2	0.342	0.340	0.646	0.647	+0.304	+0.001	pass (knife-edge)
TabPFN-2.5	0.356	0.346	0.680	0.685	+0.324	+0.005	pass
5 of 5 pass; quorum 3. ACCEPTED.							

The grid is complete: zero failed cells and zero soft skips. All five learners pass both terms against a quorum of three, so the dataset is accepted. The two terms are not comparable in size. Δ_{Joint} runs from $+0.250$ to $+0.324$, two orders of magnitude above δ , while Δ_{Aware} runs from $+0.001$ to $+0.015$ and is the term that could plausibly have failed. TabPFN v2 passes on a knife edge, its rounded Δ_{Aware} equal to δ ; excluding that learner leaves 4 of 5 and the verdict stands.

Table B.2: Per-(learner, fold) Δ_{Joint} (left) and Δ_{Aware} (right), Vietnam housing, unrounded.

Learner	Δ_{Joint} by fold					Δ_{Aware} by fold				
	0	1	2	3	4	0	1	2	3	4
CatBoost	0.2847	0.2831	0.2867	0.2924	0.2823	0.0095	0.0110	0.0056	0.0067	-0.0082
LightGBM	0.2424	0.2411	0.2501	0.2661	0.2456	0.0204	0.0188	0.0120	0.0156	0.0110
TabM	0.2989	0.2958	0.2921	0.2721	0.3137	0.0112	-0.0022	0.0056	0.0043	0.0052
TabPFN v2	0.2875	0.2998	0.3039	0.2808	0.3042	0.0063	0.0018	0.0036	0.0019	-0.0064
TabPFN-2.5	0.3103	0.3292	0.3167	0.3099	0.3097	0.0066	0.0024	0.0071	0.0102	0.0013
Pooled Δ_{Joint} : $n=25$, mean 0.2880, sd 0.0240, 25/25 positive, $t = 59.88$.										
Pooled Δ_{Aware} : $n=25$, mean 0.0065, sd 0.0068, 22/25 positive, $t = 4.75$, range $[-0.0082, +0.0204]$.										

Pooled over the 25 cells, Δ_{Joint} is 0.2880 ± 0.0240 and positive in every one ($t = 59.88$), while Δ_{Aware} is 0.0065 ± 0.0068 and positive in 22 ($t = 4.75$). The three negative Δ_{Aware} cells fall on three different learners rather than concentrating in one, so no learner's pass rests on a systematically adverse fold. This is the pattern that distinguishes this dataset from Udemy courses: the awareness term is small here, but it is consistently signed.

C Udemy courses - complete T4 grid

Table C.1: REG_TEXT_EDU_UDEMY_ACADEMY, 100 cells, one Tesla T4, ft at 10 epochs. Sources: grid_gpu_light_cat_tabm.csv, grid_gpu_tabpfn.csv.

Learner	Struct.	Text	Frozen	TAR	Δ_{Joint}	Δ_{Aware}	Verdict
CatBoost	0.297	0.210	0.480	0.501	+0.183	+0.021	pass
LightGBM	0.261	0.212	0.479	0.474	+0.218	-0.005	fail
TabM	0.298	0.221	0.504	0.492	+0.206	-0.012	fail
TabPFN v2	0.299	0.162	0.435	0.437	+0.136	+0.002	pass
TabPFN-2.5	0.311	0.211	0.452	0.464	+0.141	+0.012	pass
3 of 5 pass; quorum 3. ACCEPTED.							

Three of five learners pass, exactly the quorum, so the dataset is accepted. The split falls entirely on the awareness term: Δ_{Joint} is large and positive for every learner, from +0.136 to +0.218, while Δ_{Aware} is negative on LightGBM and on TabM. CatBoost carries the widest awareness margin at +0.021 and TabPFN v2 the narrowest at +0.002. An acceptance resting on the minimum quorum with margins this thin is one we report with the qualification it deserves; the per-fold table below.

Table C.2: Per-(learner, fold) Δ_{Joint} (left) and Δ_{Aware} (right), Udemy courses, unrounded.

Learner	Δ_{Joint} by fold					Δ_{Aware} by fold				
	0	1	2	3	4	0	1	2	3	4
CatBoost	0.1801	0.2253	0.1411	0.2115	0.1533	0.0029	0.0423	0.0131	0.0069	0.0418
LightGBM	0.2248	0.2431	0.1836	0.2238	0.2054	-0.0169	-0.0005	0.0288	-0.0479	0.0077
TabM	0.2012	0.2605	0.2252	0.1889	0.1231	-0.0221	-0.0148	-0.0189	-0.0099	0.0067
TabPFN v2	0.1318	0.1498	0.0981	0.1429	0.1583	-0.0117	0.0178	0.0068	0.0055	-0.0100
TabPFN-2.5	0.1301	0.1864	0.0863	0.1433	0.1554	0.0208	-0.0002	0.0133	0.0094	0.0193
Pooled Δ_{Joint} : $n=25$, mean 0.1749, sd 0.0459, 25/25 positive, $t = 19.06$.										
Pooled Δ_{Aware} : $n=25$, mean 0.0036, sd 0.0202, 15/25 positive, $t = 0.89$ (not significant), range $[-0.0479, +0.0423]$.										

The contrast with Appendix B is the point of this table. Δ_{Joint} is positive in all 25 cells and pools to 0.1749 ± 0.0459 ($t = 19.06$). Δ_{Aware} changes sign within a single learner across folds - LightGBM spans -0.0479 to +0.0288 - and pools to 0.0036 ± 0.0202 , not significant at $t = 0.89$. A per-learner awareness verdict on this dataset is therefore a statement about a margin narrower than the fold-to-fold noise, which is why only the dataset-level verdict should be cited.

D Rejected candidates - complete grids

Table D.1: Board games (melissamonfared/board-games), 100 cells on one T4, measured with the *maximal* structured block restored.

Learner	Struct.	Text	Frozen	TAR	Δ_{Joint}	Δ_{Aware}	Verdict
CatBoost	0.576	0.490	0.623	0.623	+0.047	0.000	fail
LightGBM	0.569	0.477	0.616	0.619	+0.047	+0.003	pass
TabM	0.576	0.504	0.628	0.629	+0.052	+0.001	knife-edge (counted fail)
TabPFN v2	0.585	0.504	0.640	0.639	+0.055	-0.001	fail
TabPFN-2.5	0.586	0.514	0.645	0.645	+0.059	0.000	fail
1 of 5 counted honestly (2 of 5 nominally); quorum 3. REJECTED.							
Pooled Δ_{Joint} : mean 0.0523, sd 0.0072, 25/25 positive, $t = 36.21$.							
Pooled Δ_{Aware} : mean 0.0003, sd 0.0058, 10/25 positive, $t = 0.25$, range $[-0.0124, +0.0163]$.							

Δ_{Joint} is consistently positive but Δ_{Aware} is not. Every learner clears the joint term, which pools to 0.0523 ± 0.0072 and is positive in all 25 cells, but the awareness term pools to 0.0003 ± 0.0058 with

only 10 of 25 cells positive and $t = 0.25$ - indistinguishable from zero. TabM's Δ_{Aware} of $+0.001$ equals δ after rounding and is counted as a fail, which is why the table reads 1 of 5 counted honestly against 2 of 5 nominally. Either count falls short of the quorum. This candidate is the clearest instance of the pattern in Figure A.1: a real joint gain that the awareness term nonetheless rejects.

Table D.2: Anime popularity (wiltheman/anime-data-set-for-ml, 10,000 rows, target popularity, text = genres, themes, title_english; members and scored_by dropped as leakage). 80 cells on one T4.

Learner	Struct.	Text	Frozen	TAR	Δ_{Joint}	Δ_{Aware}	Verdict
CatBoost	0.821	0.441	0.852	0.851	+0.031	-0.001	fail
LightGBM	0.816	0.435	0.847	0.847	+0.031	0.000	fail
TabM	0.816	0.442	0.853	0.851	+0.037	-0.002	fail
TabPFN v2	0.826	0.455	0.859	0.857	+0.033	-0.002	fail
TabPFN-2.5	not run						non-pass

0 of 5; quorum 3. REJECTED.
Pooled Δ_{Joint} : $n=20$, mean 0.0329, sd 0.0066, 20/20 positive, $t = 22.31$.
Pooled Δ_{Aware} : $n=20$, mean -0.0012, sd 0.0026, 7/20 positive, $t = -2.15$.

TabPFN-2.5 was not run, so its cell counts as a non-pass and the quorum denominator stays at five. The dataset fails at 0 of 5. As with board games the joint term is comfortable, pooling to 0.0329 ± 0.0066 and positive in all 20 measured cells. What sets this candidate apart is the direction of the awareness term: Δ_{Aware} pools to -0.0012 at $t = -2.15$, so the tuned encoder is significantly *worse* than the frozen one. This is the only candidate we gridded where adapting the encoder to the target actively hurt.

Table D.3: Metacritic PC games - rejected at the target.

Quantity	Value
Source rows	12,624
Rows with sentinel Metacritic = 0 ("never reviewed")	10,357 (82%)
Scored rows after repair	2,267 (20-96, 71 distinct, $ z _{\text{max}} = 4.67$)
Surviving text columns	1 (ResponseName, a title - a proper noun, not free text)
Structured columns	15, dominated by RecommendationCount (importance 0.62)
Grid completeness	52 rows, 3 frozen states, no ft state
Fold-0 Δ_{Aware} screen	-0.014 (LightGBM), -0.006 (CatBoost) - wrong sign on both

The rejection happened before a verdict was worth computing. A sentinel Metacritic score of zero, meaning never reviewed, covers 82% of rows, so the nominal regression is really a has-a-score indicator; repairing it leaves 2,267 scored rows. The only surviving text column is a title, a proper noun rather than free text, and the fold-0 Δ_{Aware} screen comes out the wrong sign on both learners tried. The partial grid is reported for completeness, and no verdict is claimed from it.

E MTG card prices - complete T4 grid, borderline

Table E.1: REG_TEXT_GAMES_MTG_CARD_PRICES (douglascampospire/mtg-all-cards, 28,507 cards, target $\log_{10}(\text{USD})$, text = CARD_TEXT, TYPE), 100 cells on one T4, ft at 10 epochs. Sources: grid_gpu_light_cat_tabm.csv, grid_gpu_tabpfm.csv.

Learner	Struct.	Text	Frozen	TAR	Δ_{Joint}	Δ_{Aware}	Verdict
CatBoost	0.525	0.292	0.586	0.587	+0.061	+0.001	knife-edge
LightGBM	0.518	0.271	0.572	0.573	+0.054	+0.001	knife-edge
TabM	0.504	0.305	0.578	0.580	+0.074	+0.002	pass
TabPFN v2	0.520	0.269	0.573	0.579	+0.053	+0.006	pass
TabPFN-2.5	0.523	0.310	0.592	0.591	+0.069	-0.001	fail

4 of 5 nominally, 2 of 5 without the knife-edges; quorum 3. BORDERLINE.
Pooled Δ_{Joint} : $n=25$, mean 0.0620, sd 0.0108, 25/25 positive, $t = 28.85$.
Pooled Δ_{Aware} : $n=25$, mean 0.0018, sd 0.0049, 18/25 positive, $t = 1.87$, range $[-0.0122, +0.0096]$.

The joint term is solid: Δ_{Joint} is positive in all 25 cells and pools to 0.0620 ± 0.0108 ($t = 28.85$). The awareness term decides the verdict and does not settle it. The reference implementation passes four learners, but CatBoost and LightGBM clear δ only on a knife-edge: their rounded Δ_{Aware} equals δ , with unrounded values of +0.0009 and +0.0011. Without them, the count is 2 of 5, below quorum. This is the only candidate whose verdict depends on how knife-edges are counted, so we report it as borderline and do not count it as an accepted dataset. The target keeps $|z|_{\text{max}} = 5.36$ after the log transform, which is warned about but never clipped, per the reference protocol.

F Validity evidence: the two invalid shortcuts

We assessed two plausible ways to cut curation cost, both of which were measurably invalid in that both collapsed the quantity being measured. These were: **(1) Deleting structured columns by name.** A filter that drops columns matching year or time removes Year Published and Play Time from a board-games candidate. These are not identifiers, but two of its strongest structured predictors, so removing them allows the text to proxy for the missing information and creates a joint gain that disappears when the columns are restored. The unstructured baseline never moves, so this is a controlled ablation rather than a correlation. Name patterns are therefore used only to reject *text* candidates. **(2) Cheap screening of Δ_{Aware} .** A cheap screening procedure is valid only if reducing computational cost does not alter the quantity being measured. That holds for Δ_{Joint} (frozen encoders, so a fold subset is unbiased) and fails for Δ_{Aware} on two independent axes. An under-trained adapter drives TAR toward frozen by construction, and a single fold cannot resolve a threshold sitting deep inside the per-(learner, fold) noise band. Screening on one fold cost two full grids.

Table F.1: **Axis 1 - structured-column deletion manufactures Δ_{Joint} .** Board games, LightGBM, fold 0, target Complexity Average, text = Mechanics + Name.

Specification	Struct.	Text	Frozen	Δ_{Joint}
auto-spec (Year Published, Play Time deleted)	0.584	0.502	0.623	+0.0388
Year Published restored	0.613	0.502	0.636	+0.0229
full structured block restored	0.684	0.502	0.684	-0.0005
same, semantically natural target (Rating Average)	0.567	0.335	0.567	+0.0006

Columns are restored one at a time. The unstructured baseline never moves from 0.502, so the entire effect sits on the structured baseline - which is what makes this a controlled ablation rather than a correlation. Deleting Year Published and Play Time yields $\Delta_{\text{Joint}} = +0.0388$; restoring Year Published alone cuts that to +0.0229; restoring the full block drives it to -0.0005. The apparent joint gain was an artefact of the deletion at every step, and a semantically natural target behaves the same way at +0.0006. Name patterns are therefore used in this pipeline only to reject *text* candidates, never to drop structured columns.

Table F.2: **Axis 2a - the epoch budget.** Identical cells, Udemy, fold 0, at two and at ten epochs.

Learner (fold 0)	Frozen	TAR @ 2 ep	Δ_{Aware} @ 2 ep	TAR @ 10 ep	Δ_{Aware} @ 10 ep
LightGBM	0.4957	0.5056	+0.0099	0.5279	+0.0322
CatBoost	0.5177	0.5282	+0.0105	0.5299	+0.0122
At 2 epochs the full Udemy sweep returned 1 of 5 - an artefact of the budget, not of the dataset.					

An under-trained LoRA adapter leaves TAR close to frozen, collapsing Δ_{Aware} toward zero by construction. Raising the budget from 2 to 10 epochs more than triples LightGBM’s Δ_{Aware} , from +0.0099 to +0.0322, and moves CatBoost’s from +0.0105 to +0.0122. At 2 epochs the full Udemy sweep returned 1 of 5 - an artefact of the budget, not a property of the dataset. A *fail* at a low budget therefore proves nothing. Only a pass is informative. The same asymmetry is why the 10-epoch budget used throughout can suppress an acceptance but cannot manufacture one.

Table F.3: **Axis 2b - the fold count.** Fold-0 Δ_{Aware} against the five-fold mean on the two gridded rejections.

Learner	Board games			Anime		
	fold 0	5-fold	sign	fold 0	5-fold	sign
CatBoost	+0.0074	−0.0003	flip	+0.0018	−0.0008	flip
LightGBM	+0.0163	+0.0021	—	+0.0034	−0.0005	flip
TabM	+0.0052	+0.0010	—	+0.0010	−0.0016	flip
TabPFN v2	+0.0010	−0.0015	flip	−0.0041	−0.0020	—
TabPFN-2.5	+0.0064	+0.0001	—	not run		
Per-(learner, fold) spread: board games $\sigma = 0.0058$ over $[-0.0124, +0.0163]$; anime $\sigma = 0.0026$ over $[-0.0066, +0.0034]$. The threshold $\delta = 0.001$ sits inside both bands, so one fold cannot resolve the criterion. Screening on one fold cost two full grids.						

Every board-games learner drops between the two columns and two change sign. Anime’s positive fold-0 reading reverses entirely, with three of its four measured learners flipping. The per-(learner, fold) spread quoted at the foot of the table puts $\delta = 0.001$ well inside both noise bands, so a single fold cannot resolve the criterion in either direction. We screened on one fold before establishing this, and it cost two full grids.

G Screen-time rejections

Table G.1: Candidates rejected before a full grid was spent, each with the measured quantity that condemns it.

Candidate	Reason (with the measured quantity that condemns it)
steam-games-dataset-steampyspyauto-target	was appid, a primary key; no free text besides proper nouns
gog-com-video-games	auto-target was a date; re-screened $\Delta_{\text{Joint}} = -0.0003$, text $R^2 = 0.996$ (the text <i>was</i> the release dates)
popular-video-games-1980-2023	12,261 rows; no numeric column qualified as a target
lego-sets	12,261 rows are 744 products replicated across 21 countries — near-duplicates straddle train and test
top-women-chess-players	$\Delta_{\text{Joint}} = +0.024$ inside the ± 0.015 band, text $R^2 = 0.074$ (text nearly inert)
kerala-bevco-liquor-price	auto-target was S1 No; with the corrected target, structured $R^2 = 1.000$ (deterministic)
google-play-store-apps	target derived from its own text
daraz-perfumes	every state negative ($-0.130 / -0.009 / -0.045$); a delta between them is meaningless
global-grocery-nutrition-2023	$\Delta_{\text{Joint}} = +0.0005$ with structured $R^2 = 0.969$ — saturated, no headroom
global-movies-1950-2026	target <code>roi_pct</code> = revenue/budget, an arithmetic identity over two structured columns; structured $R^2 = 0.997$
steam-games-dataset	no numeric column qualified as a target
ultimate-games-15k	auto-target was <code>serial_no</code> ; $\Delta_{\text{Joint}} = -0.0027$ against structured $R^2 = 0.983$
horror-movies-profits	Δ_{Joint} healthy ($+0.055 / +0.093$) but fold-0 $\Delta_{\text{Aware}} -0.001 / -0.013$
vgsales-userscore	every delta at noise scale: $\Delta_{\text{Joint}} +0.001/+0.004$, $\Delta_{\text{Aware}} +0.001/+0.002$

Two classes recur. The first is a junk target chosen automatically: `appid`, `S1 No` and `serial_no` are primary keys or serial numbers, and one candidate had a release date selected as its target. The second is a saturated or degenerate baseline against which no text signal can be demonstrated, with structured R^2 reaching 0.969, 0.983, 0.997 and 1.000 on four separate candidates. Both classes are cheap to detect from a profile of the target and the structured block, and neither needs a grid to rule out - which is what makes the profiling stage of the funnel worth its cost.

H Harness validation against the benchmark’s own anchor

Table H.1: Anchor dataset MUL_TEXT_PRODUCT_SENTIMENT ($N = 5,091$, 4 classes, 1 text + 1 structured column); 36-run frozen grid.

Check	Result
Published 56×10 pass matrix re-derived from released scores	0 mismatched cells
Verdict function on a known accepted dataset	accepted, 5 of 5
Verdict function on a known rejected dataset	rejected, 0 of 5
<code>no_text</code> (encoder-free) vs published, mean absolute difference	0.0000
LightGBM <code>no_text</code> , fold 0	0.83454303717305 (identical)
<code>text_only</code> vs published, mean signed difference	-0.0000 (range -0.046 to $+0.038$)
<code>all</code> vs published, mean signed difference	$+0.0058$ (max $+0.0235$)
Frozen-embedding cache, cached vs uncached score	0.8618478948517495, bit-identical

These checks ran before any candidate number was trusted. The published 56×10 pass matrix re-derives from the released scores with zero mismatched cells, and the verdict function returns the published outcome on both a known accepted and a known rejected dataset. The encoder-free `no_text` condition reproduces published scores exactly - to fourteen decimal places on the

LightGBM fold-0 cell - which pins loading, curation, splitting and seeding. The encoder-dependent conditions agree less tightly, all carrying a $+0.0058$ mean signed difference. That gap is the expected consequence of a different embedding environment, not of a different protocol, and it is why the next table validates Δ_{Joint} itself rather than the raw scores it is built from.

Table H.2: Δ_{Joint} - the quantity the criterion consumes - reproduced on the anchor.

Learner	Δ_{Joint} (ours)	Δ_{Joint} (published)	Sign agrees
CatBoost	0.075	0.073	yes
LightGBM	0.058	0.046	yes
TabM	0.085	0.086	yes
TabPFN v2	0.076	0.079	yes

The deltas agree within 0.012 across four learners with every sign agreeing, and three of the four agree within 0.003. The single wider case is LightGBM, at 0.058 against a published 0.046. This reproduction is the calibration behind the ± 0.015 fold-noise band used throughout the paper: a candidate whose Δ_{Joint} falls inside that band is not distinguishable from measurement noise, and is screened out rather than gridded.